

Choosing and Combining Biological Foundation Models for Sequence Prediction

Literature Review

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The question behind biological sequence fusion

Biological foundation models offer a way to predict properties of DNA, RNA, and proteins from their sequences. The question for this project is when combining their representations improves a biologically defined prediction. BioLangFusion and IsoFormer provide evidence that combining sequence representations can help, but they also show that the benefit depends on the task, the inputs, and the method of combination. Their results motivate a closer examination of what an added representation contributes, rather than an assumption that more modalities necessarily produce a better model (Mollaysa et al. 2–4; Garau-Luis et al. 5–9).

This review develops that question through eight selected papers. Nucleotide Transformer and ESM-2 establish the sequence-model background; BioLangFusion and IsoFormer provide the main fusion comparison; LucaOne, Central Dogma Transformer, and Coupled Mamba show other ways to integrate biological information. Research on multimodal representation alignment helps interpret what similarities between model outputs can establish. The discussion begins with the biology needed to distinguish the inputs, then examines the methods and experimental evidence, and ends with a proposed investigation. The selected studies support a focused comparison, not a claim to exhaustive coverage of the field.

Biological information and learned representations

DNA and RNA are chains of nucleotides, whereas proteins are chains of amino acids. During transcription, RNA polymerase uses DNA as a template to build an RNA molecule. During translation, a ribosome reads a protein-coding mRNA and joins amino acids in the specified order, with transfer RNAs bringing those amino acids. The message supplies instructions; the amino acids supply the material of the protein. This distinction matters because

a computational model receives a sequence describing one of these molecules, not the cellular machinery that produces or uses it (Alberts et al., *Essential*, ch. 7; Nelson et al., secs. 3.1–3.2).

Within a coding region, groups of three nucleotides, called codons, specify amino acids or termination signals in a particular reading frame. Several codons can specify the same amino acid. Consequently, distinct coding sequences can translate into the same protein sequence, and that protein sequence alone does not identify the original synonymous codons. This is a concrete reason to keep the nucleotide and protein views distinct. It also means that translating a supplied coding sequence does not add an independent measurement of the cell (Nelson et al., sec. 27.1). A protein model may still make useful patterns easier to access because of what it learned during pretraining; this is a possible explanation of fusion benefit that must be evaluated against the task results.

The genomic region and transcript region supplied to a model also matter. Regulatory DNA helps control transcription, while RNA processing removes introns and joins exons. Alternative splicing can produce different transcript isoforms from the same gene. Mature mRNAs can also contain untranslated regions that influence localization, translation, and persistence. A coding sequence therefore differs from a full transcript, and both differ from a genomic window containing nearby regulatory DNA. Replacing U with T in an RNA string does not recover the regulatory sequence around its gene (Alberts et al., *Molecular* 334, 339–41, 447, 455–60). These distinctions become central when comparing BioLangFusion with IsoFormer.

A prediction target must be identified just as carefully as an input. RNA abundance depends on both synthesis and degradation: a change in either can alter how much RNA is present. RNA stability concerns persistence or decay, whereas transcript-expression measurements concern abundance under specified conditions. An expression predictor therefore

does not automatically identify the process responsible for that abundance (Nelson et al., sec. 26.2). In the studies reviewed here, the same broad language of biological prediction covers several different quantities. Their scores cannot be interpreted without their targets and measurements.

A sequence model converts input units, or tokens, into numerical representations called embeddings. An encoder produces these features; a prediction head uses them to estimate a target. In a transformer, attention allows a position's representation to depend on other positions. Pretraining teaches the model through an initial objective, such as reconstructing selected sequence tokens. A later prediction task uses labeled examples to train a head or adapt parts of the model. This separates learning general sequence patterns from learning the particular relationship being evaluated (Dalla-Torre et al. 288, Methods, “Models” and “Fine-tuning”).

Nucleotide Transformer illustrates why the representation choice itself deserves attention. Its benchmarks cover tasks including promoters, splice sites, and regulatory annotations. The authors find that useful information depends on the layer examined: the final layer was not the best representation in their probing comparison. They also report stronger results when training added weights that rescale features, together with prediction heads, while keeping the original pretrained weights fixed. Thus, a model name does not specify a single, universally useful feature set. Which layer is used and how it is adapted can affect the outcome of a later fusion comparison (Dalla-Torre et al. 288, 294–95, Methods, “Fine-tuning”).

ESM-2 provides the protein-sequence counterpart. Lin et al. investigate the relationship between protein language modeling and structure prediction. The accompanying author documentation distinguishes ESM-2 embeddings from ESMFold, the system that uses language-model representations to predict a protein's structure. Supplying ESM-2 features to an expression

predictor therefore does not mean supplying predicted atomic coordinates or measured protein activity (Lin et al.; Meta Fundamental AI Research Protein Team). Together, the nucleotide and protein models motivate testing reusable representations while leaving their contribution to any particular prediction to be established experimentally.

What sequence fusion has demonstrated

BioLangFusion begins with mRNA sequences and derives corresponding DNA and protein views. It uses Nucleotide Transformer v2, RNA-FM, and ESM-2 to obtain representations, then brings the streams to a common positional resolution based on the codon-to-amino-acid relationship. This establishes a way to combine corresponding positions. It does not, by itself, establish that the numerical features have the same meaning or contain independent biological information. Its biological interpretation depends on the supplied sequence, coding frame, and handling of positions outside the coding region (Mollaysa et al. 2–3, 8).

The paper compares three ways to combine the representations. Concatenation places their features side by side. Modality attention learns how strongly to weight the streams. Cross-attention allows one stream to draw information from another before prediction. These fused features feed a convolutional prediction head that combines features across nearby positions. The study reuses pretrained models and trains downstream components; its claim of avoiding additional pretraining should not be read as a claim that the system requires no training. The paper's wording does not fully specify every encoder's gradient setting, so a strict frozen-encoder characterization would need clarification (Mollaysa et al. 2–4, 9–11).

The main results support a conditional benefit. On transcript stability, modality attention with entropy regularization reaches a Spearman correlation of 0.563, compared with 0.553 for RNA-FM alone. Spearman correlation measures agreement in the ordering of predicted and

measured values. The other fusion methods reach 0.539 for concatenation and 0.550 for cross-attention, both below RNA-FM on this task. Across the five reported tasks, at least one fusion method exceeds the strongest listed single encoder, but the successful method is not always the same. The main table does not provide repeated-run uncertainty, limiting what can be concluded about small differences (Mollaysa et al. 4, 8).

These results make BioLangFusion a useful starting point, but they leave the source of its benefit open. Its derived protein view can expose patterns learned from protein data without adding a new observation of the cell. A comparison between two differently pretrained nucleotide encoders could help determine whether some benefit also arises from combining models within one modality. That is a proposed control motivated by the method, not a result reported by the paper. The distinction matters for deciding whether a gain should be attributed to biological sequence views, pretraining differences, or the additional prediction components.

IsoFormer adds genomic context and explicitly describes joint fine-tuning. It combines DNA around a transcript's transcription start site, transcript sequence including untranslated regions, and the corresponding protein when available. Its target is transcript abundance across 30 tissues in GTEx, using measurements averaged across individuals and transformed for prediction. Unlike BioLangFusion's DNA view derived from an RNA string, its genomic input supplies additional sequence context. Successive cross-attention operations exchange information among streams, and the system jointly fine-tunes the pretrained encoders and downstream components (Garau-Luis et al. 4–6, 16–17).

IsoFormer's modality comparison reports R^2 values of 0.36 ± 0.03 for RNA alone, 0.39 ± 0.01 for DNA plus RNA, and 0.43 ± 0.01 for all three inputs. R^2 evaluates prediction error relative to a mean-target baseline; these values are not percentages of transcripts correctly

classified. The reported variation is the standard deviation across five seeds. These runs share a gene-separated test set, so their variation reflects training and validation choices within that dataset. Replacing the DNA encoder with Enformer raises the full system's R^2 to 0.53 ± 0.01 , but the DNA-plus-RNA version already reaches 0.52 ± 0.01 . A different aggregation method also reaches 0.53 ± 0.01 . The evidence supports the combined system on this expression task, while showing that protein inclusion and cross-attention are not uniformly decisive (Garau-Luis et al. 6–9, 18).

The two papers therefore support complementary conclusions rather than a ranking. BioLangFusion tests relatively simple combinations of representations derived from related sequence views across several targets. IsoFormer tests an integrated, jointly adapted system with additional genomic context for tissue-average transcript expression. Their datasets, targets, and metrics differ. Even within IsoFormer, changing the DNA encoder changes model architecture, pretraining, and context length together. Its improved result cannot be assigned to any one of those changes from that comparison alone (Mollaysa et al. 4; Garau-Luis et al. 5–9, 17–18).

Other ways to integrate biological information

LucaOne moves integration into pretraining. It uses a shared transformer for nucleic-acid and protein sequences, combining masked-token objectives with annotation and structure tasks. The authors report that explicit DNA–protein pairing was not supplied during pretraining. This offers a different approach from joining encoders that were pretrained separately: the model's parameters are shaped by both sequence types before the downstream task (He et al. 943, 948–49).

Its central-dogma experiment tests whether a DNA sequence and a protein sequence correspond. Frozen representations are pooled, concatenated, and given to a supervised classifier.

The classifier receives labeled examples of matching and nonmatching pairs; it does not generate a protein from DNA or reproduce transcription and translation. LucaOne performs better than the tested separately pretrained alternatives in this comparison, supporting the usefulness of its shared pretraining for the pairing task. However, the displayed comparison combines validation and test results, and the described random pair split does not establish separation by gene. These features limit claims about generalization to unfamiliar biological families or about unsupervised acquisition of the entire DNA-to-protein relationship (He et al. 944–46, 950).

Central Dogma Transformer instead builds a direction of information flow into its architecture. In the inspected version, RNA representations query DNA representations, and protein representations then query the fused RNA stream. Its RNA input consists of static gene-token embeddings, rather than sample-specific transcript sequences or expression values; its RNA and protein matrices remain the same across samples. The prediction target is the expression effect of silencing an enhancer in a K562 cell-line dataset. Ota reports a validation Pearson correlation of 0.503 across 996 enhancer–gene pairs. This is evidence about prediction in that validation setting, not an independent test of a general cellular simulator (Ota 12–13, 29–33).

The directional design gives a biological hypothesis a computational form, but it does not establish that the attention operations recover the corresponding cellular mechanisms. Ota's interpretation of a particular locus draws on prediction sensitivity and existing biological annotations, while acknowledging that prospective perturbation is needed to assess the proposed mechanism. The fixed RNA and protein inputs also leave open how the method would behave with varying cellular states. This paper is therefore most useful here as an example of how

architecture can guide hypothesis generation, with mechanistic conclusions still requiring additional evidence (Ota 22–25).

Coupled Mamba broadens the meaning of multimodal biological input further. Its coupled state-space method exchanges information through evolving hidden states, offering an architectural alternative to attention-based fusion. The published NeurIPS paper includes breast-cancer subtype classification using measured mRNA expression, DNA methylation, and microRNA expression. It reports 88.1% accuracy, compared with 87.5% for MM-Dynamics. These are measurements associated with samples, rather than DNA, RNA, and protein sequence embeddings. The result supports studying the architecture for biological data, but it does not demonstrate the same benefit on sequence-based stability or expression prediction. Nor does subtype classification establish prospective clinical utility (Li et al. 4–8).

Together, these studies show why a review must identify what is being integrated. Shared pretraining, fusion of sequence representations, and integration of molecular measurements address different problems. Their results can motivate design alternatives, but moving a method between these settings requires a new comparison with the appropriate inputs and biological target.

Alignment and the proposed research contribution

The word alignment can obscure several different questions. BioLangFusion's codon correspondence concerns matching biological positions. Representation alignment concerns whether encoders organize corresponding examples similarly in their numerical feature spaces. Fusion concerns whether combining those features improves a prediction. Establishing the first does not settle the second or third. This distinction explains why biologically corresponding

sequences may still produce differently organized representations, and why either similarity or difference needs interpretation through a task (Mollaysa et al. 2; Tjandrasuwita et al. 3–5).

Tjandrasuwita et al. investigate these relationships using controlled synthetic data and nonbiological multimodal benchmarks. They find that alignment depends on shared versus modality-specific task information. Its relationship with performance varies: some settings show strong positive associations, while others show weak or negative ones. Experiments that vary the strength of an alignment objective also show task-dependent effects. This provides a reason to measure representational similarity and predictive benefit separately. It does not establish the corresponding relationship for DNA, RNA, and protein models, which remains an empirical question for this project (Tjandrasuwita et al. 4–9).

A defensible next investigation is therefore to ask whether a measured relationship between two sequence representations helps explain when their combination improves prediction. The first comparison should use one defined biological target, the same examples and data split, and comparable prediction heads and tuning effort. Single-encoder and combined-encoder results would establish the predictive difference; representation measurements would then help describe it. Simple sequence features and combinations of encoders from the same modality could test alternative explanations. These are proposed comparisons, not claims that the reviewed studies exhaust the available controls or that the proposed work is unprecedented.

The biological interpretation should remain tied to the endpoint. A stability model could help prioritize sequence changes for an experiment measuring RNA persistence; an expression model could help prioritize questions about transcript abundance; an enhancer-effect predictor could suggest regulatory relationships to test by perturbation. These are possible uses motivated by the reviewed targets. A useful prediction does not alone establish its causal explanation, and a

disease-related classification result does not demonstrate treatment value. The contribution sought here is a clearer account of when combining representations adds predictive value and what biological interpretation that added value can support.

The selected literature establishes that biological representations can be useful and that some combinations improve specific predictions. It also shows that the gains depend on information supplied to the model, pretraining and adaptation, and the evaluation task. The project can build on that evidence by testing a narrow comparison and explaining its outcome in those terms.

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Preparation and AI Assistance

Chase Di Maria Breisinger used Codex (OpenAI) to prepare and revise this literature review. He supplied the research canvas, original and AI-refined Obsidian notes, three biology textbooks, the professor's annotated review, and guidance from Practical Research. He selected the narrower source scope and requested clearer biological interpretation, source checking, and a connected account of the research. His notes informed the distinction between mRNA instructions and amino-acid building blocks, and the emphasis on explaining biological objects before model operations. This influence does not establish that he independently verified every statement in the draft.

Julian McKinley's contribution to the preparation of this same review included material produced with Claude (Anthropic). His disclosure described assistance with finding sources, drafting, formatting, and checking available publication information. The available record does not establish which sources Julian personally selected or verified. The broader source expansion from that preparation is not retained in this revision. No separate student review is treated as scientific evidence or cited as another paper.

Codex generated most of the prose, source comparisons, and proposed synthesis in this revision. It inspected relevant primary-paper and textbook passages, checked publication identities, and revised the content in response to the professor's comments about organization, unclear explanations, and ambiguous figure references. The scientific discussion is limited to the eight selected core papers, three biology textbooks, and the official ESM documentation for the distinction between embeddings and structure prediction. BioLangFusion is cited as its June 2025 version 1 workshop preprint, Central Dogma Transformer as its January 2026 version 2

preprint, and Coupled Mamba as the published NeurIPS paper, which contains experiments absent from the locally retained preprint.

Source checking was targeted rather than a complete line-by-line audit of every source. For ESM-2, the final Science full text was inaccessible through the available retrieval routes; coverage was limited to the author abstract, publication metadata, and official repository documentation. Paper-supported numerical comparisons elsewhere were checked against the relevant methods and results. Parenthetical locators use printed journal pages where available and internal page numbers for conference/preprint PDFs; reflowed textbooks are cited by chapter or section. Statements about proposed comparisons and laboratory uses are interpretations or proposals, not completed experiments.

No experiments or notebooks were rerun for the review, and independent verification of the final prose, references, and ethical considerations by both authors is not yet documented. The ethical implications of research automation have not been fully examined. Chase intends to revisit the draft as he reads the papers and develops his own Obsidian analyses. The cover retains the requested assignment date of 4 October 2026; this revision was prepared on 9 October 2026.

Preparation guidance: Leedy, Paul D., and Jeanne Ellis Ormrod, with Laura Ruth Johnson. *Practical Research: Planning and Design*. 12th ed., Global ed., Pearson, 2021, pp. 98–102. The chapter's guidance on evaluating, organizing, and synthesizing literature informed this revision's structure; it is not evidence for its biological or model-performance claims.